

3. CRITICALITY SAFETY ANALYSIS

3.1 MODEL DESCRIPTION

Transportation regulations [9] require evaluation of a single package in isolation under NCT and HAC and package arrays under NCT and HAC analysis. This evaluation considered increased enrichments for all of these configurations.

The nonproprietary DN30 SARP [2] does not provide sample inputs or sufficient details to identically replicate the calculational models. Likewise, the UX30 SARP [7] does not provide sample inputs. The models for this evaluation were based on the HAC models developed for ORNL's *Assessment of Existing Transportation Packages for Use with HALEU* (ORNL/TM-2020/1725) [1]. Key aspects of the models include:

- The 30B cylinder volume is maximized by using the minimum wall thickness and the maximum cylinder dimensions with the cylinder modeled as carbon steel (SCALE standard material *carbonsteel*).
- UF_6 is modeled with a purity of 99.5 wt.% UF_6 containing 0.5 wt.% HF (H/U of 0.88), with varying densities.
- Overpacks are typically made of two thin metal shells with a type of energy-absorbing insulating foam material in between and can be cylindrical or rectangular in shape. The overpack is assumed to collapse into close proximity to the 30B cylinder wall (no foam material) for modeling purposes. As in the DN-30 analysis, only the overpack's stainless-steel walls were modeled in this analysis, resulting in a combined steel thickness of 0.5 cm as a right circular cylinder around the 30B cylinder (SCALE standard material *SS304*). This generic overpack model of a close-fitting right circular cylinder does not credit a specific overpack design with the presence of the inner foam-like material or the full spacing provided by the overpack. The exact overpack shape, shell thicknesses, and foam-like material composition are expected to vary between manufacturers/owners, and the foam-like material composition may be proprietary.
 - Modeling conservatisms include removing all of the foam material and the spacing it provides between 30B cylinders. Adding foam material and/or crediting the spacing provided would decrease the interaction between cylinders.

A cutaway rendering of a single 30B cylinder with a generic overpack is shown in Figure 3. Consistent with the DN-30 analysis, single package models do not include HUR. The dark blue region on the interior of the cylinder is the UF_6 , the aqua colored area is the carbon steel wall of the cylinder, the red cylinder outline is the overpack, the gray indicates the gap between the 30B cylinder and the overpack (modeled as void here), and the yellow region is the full water reflection outside the overpack.

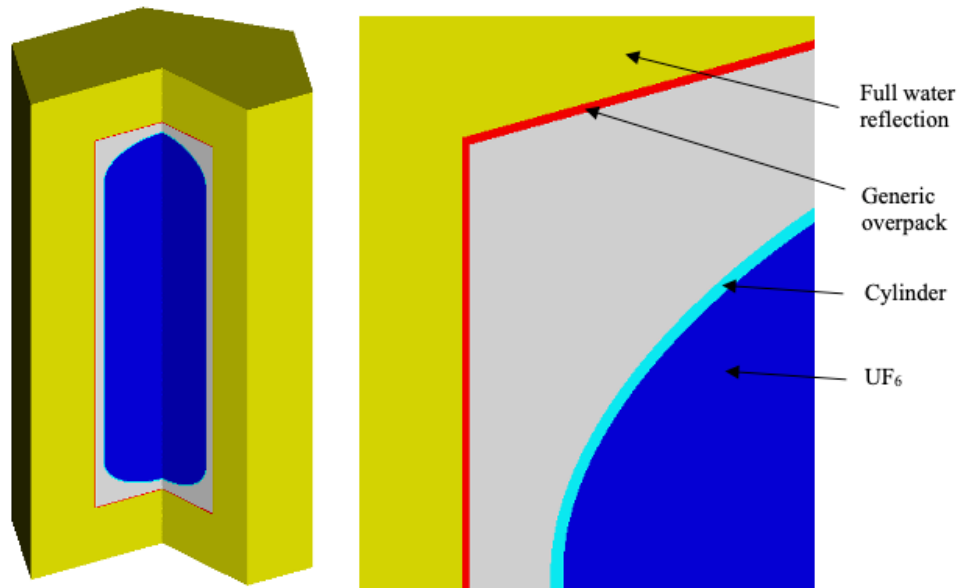


Figure 3. KENO-VI model of a single 30B cylinder with overpack and full water reflection.

The consideration of HUR in the 30B cylinder model has evolved over the years. The 1991 analysis supporting the current UX30 safety basis [15] does not specifically model any HUR in the cylinders; the DN-30 analysis considers HUR in the array configurations. Recently published papers on the nature of these residues are listed and described below:

- Begue, L., Milin, M., Caplin, G., and Evo, S., “Nuclear Criticality Safety of Enriched UF_6 Cylinders,” PATRAM 2013: describes how the formation of limited hydrated complexes of UO_2F_2 ($\text{UO}_2\text{F}_2 \cdot X\text{H}_2\text{O}$) were observed in different experiments, with the H/U ratio depending on conditions in the fuel cycle facilities and during transport (varying waters of hydration, X), with a maximum H/U ratio assumed to be less than 11 [16].
- O’Connor, G., “Regulatory Criticality Safety Review of Uranium Hexafluoride Transport Package Applications,” PATRAM 2013: states that in the view of the UK Competent Authority, some of the most conservative assumptions could be relaxed (for example, a smaller maximum H/U ratio) [17].
- Rezgui, S. and Hilbert, F., “Criticality Analyses of Enriched Uranium-Hexafluoride Containing Impurities,” PATRAM 2013: discusses the DN30B model with the complex spherical configuration and describes how the optimum HUR configuration and placement were determined [18].
- Milin, M., Rannou, J., Viaulle, L., Caplin, G., and Evo, S., “Hydration of Uranium Residues Contained in Enriched UF_6 Cylinders,” PATRAM 2016: justifies a bounding H/U ratio of 6 for uranium residues [19].

The DN-30 SARP includes these references as part of its basis for developing its complex HUR spherical arrangement. The HUR model presented in ORNL’s *Assessment of Existing Transportation Packages for Use with HALEU* [1] mimics the DN-30 model. The HUR model consists of multiple spherical layers; the inner layer is an 11.4 kg sphere of $\text{UO}_2\text{F}_2 \cdot 3\text{H}_2\text{O}$ (with a density of 6.3 g/cm^3 [20]) surrounded by a spherical layer of UF_6 , which is in turn surrounded by a spherical layer of HF (HF acid with a density of 1.15 g/cm^3 , SCALE standard material *hfacid*). The amount of HF in the spherical shell layer thickness and placement was determined through parameter studies to maximize the system k_{eff} . The HF

representing the UF₆ impurities is still included as part of the UF₆ loading as an additional conservatism. This results in additional HF being included in the model.

Of the referenced 30B cylinder package SARPS [2, 7], the HUR considerations are very different. The UX30 basis does not consider HUR in its analysis. The DN-30 basis conservatively models the HUR as an optimally shaped sphere placed to maximize interaction and system k_{eff} . The DN-30 adds conservatism with its impurity modeling. In both SARPS, the impurities (modeled as HF) are included as part of the UF₆. The DN-30 analysis adds additional HF as a spherical shell around the HUR sphere, with the shell thickness and spacing determined through parameter studies to maximize the system k_{eff} .

Given the different densities of the HUR, HF, and UF₆, in this evaluation, the HF spherical shell layer is not considered a realistic configuration. Therefore, the HUR is modeled as a single sphere with the HF mixed homogeneously throughout the UF₆. Select cases will include the HF spherical shell layer consistent with the ORNL assessment [1] models for comparison.

3.2 SINGLE PACKAGE CALCULATIONS

For single package calculations, a 30B cylinder was modeled with and without an overpack. For both configurations, the UF₆ enrichment was varied from 6 to 20 wt.%, and the UF₆ density was varied from 2.5 to 5.5 g/cm³. Consistent with the DN-30 analysis, single package models did not consider HUR. Both configurations included infinite water reflection modeled as a 30 cm thick water reflector. The overpack model includes a 5 cm gap between the cylinder and the overpack (modeled as void). Results are presented in Table 2 and Figure 4 (infinite water reflected cylinder) and Table 3 and Figure 5 (infinite water reflected cylinder with overpack). Entries shaded in blue have k_{eff} values greater than 1.0, and those shaded in gray have k_{eff} values between 0.95 and 1.0. The Monte Carlo statistical uncertainty in the k_{eff} values is not presented in the results, but all calculations in this study were run until an uncertainty of 0.05% Δk (50 pcm) was achieved.

Table 2. k_{eff} values for single 30B cylinders with infinite water reflection

UF ₆ enrichment (wt.%)	UF ₆ density (g/cm ³)			
	2.5	3.5	4.5	5.5
6	0.5049	0.5600	0.6053	0.6401
7	0.5274	0.5889	0.6393	0.6801
8	0.5458	0.6143	0.6704	0.7166
9	0.5651	0.6380	0.6992	0.7489
10	0.5825	0.6611	0.7262	0.7798
12	0.6129	0.7016	0.7763	0.8366
15	0.6533	0.7561	0.8422	0.9091
20	0.7127	0.8354	0.9348	1.0119

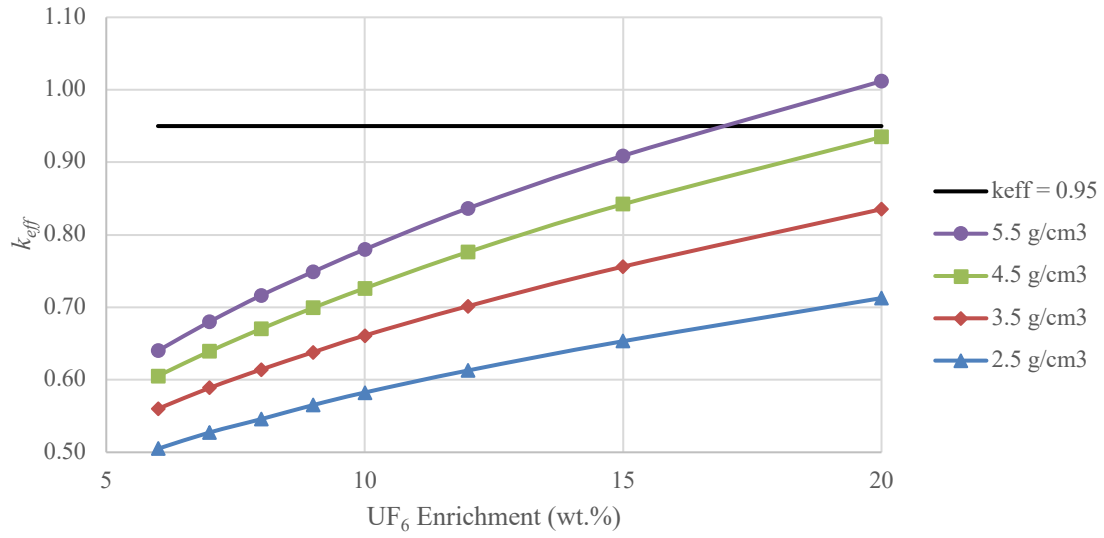


Figure 4. k_{eff} values as a function of UF_6 enrichment and UF_6 density for single 30B cylinders with infinite water reflection.

Table 3. Values for single 30B cylinders with an overpack and infinite water reflection

UF ₆ enrichment (wt.%)	UF ₆ density (g/cm ³)			
	2.5	3.5	4.5	5.5
6	0.4203	0.4928	0.5536	0.6041
7	0.4426	0.5227	0.5898	0.6461
8	0.4625	0.5492	0.6238	0.6823
9	0.4801	0.5747	0.6545	0.7170
10	0.4977	0.5996	0.6835	0.7497
12	0.5295	0.6419	0.7362	0.8079
15	0.5737	0.7005	0.8048	0.8843
20	0.6363	0.7849	0.9021	0.9889

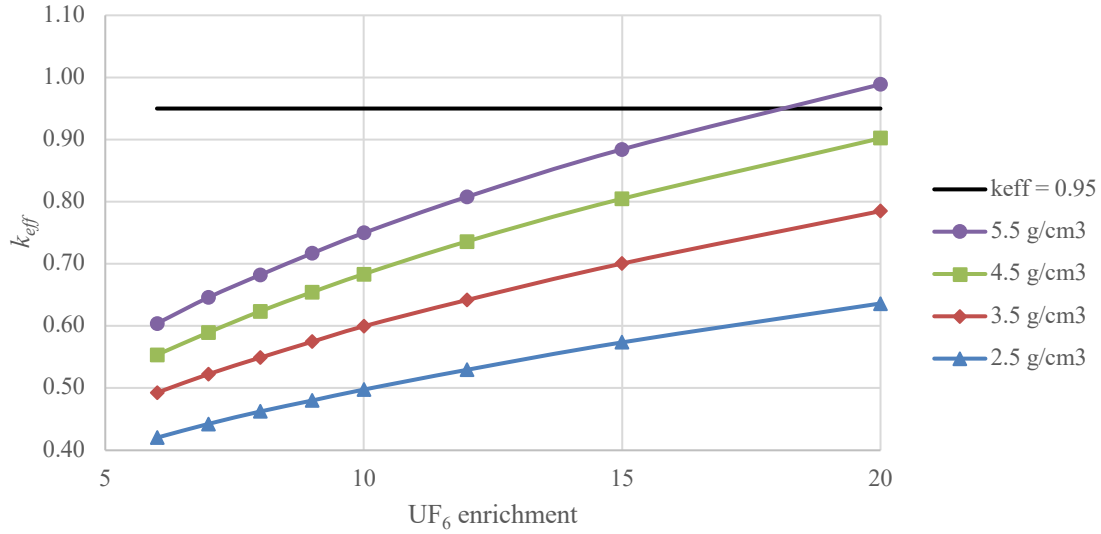


Figure 5. k_{eff} values as a function of UF₆ enrichment and UF₆ density for single 30B cylinders with an overpack and infinite water reflection.

For single 30B cylinders, k_{eff} increased with both increasing UF₆ density and UF₆ enrichment. Enrichments greater than 15 wt.%, with k_{eff} values greater than 0.95, indicate infinite arrays will not be acceptable under NCT or HAC

3.3 INFINITE ARRAY CALCULATIONS

The parametric capability within the Sampler sequence (described in Section 2.3) was used to generate models of an infinite array of 30B cylinders with variations of multiple parameters, including:

- UF₆ enrichment
- UF₆ density
- Water layer (film) thickness around the 30B cylinder (between the cylinder and the overpack)
- Density of water film

The enrichment was varied from 6 to 10 wt.% ²³⁵U, and the UF₆ density was varied between 2 and 5.5 g/cm³. As with the single package models, the infinite arrays did not consider HUR. The enrichment range was determined based on scoping calculations that indicated an infinite array of 30B cylinders achieved supercriticality (k_{eff} values greater than 1.0) within this range; higher enrichments were thus not studied in this phase of the work. This range of UF₆ densities spans the material density from liquid to full theoretical density as a solid. The water film thickness around the 30B was varied from 0 to 5 cm, with densities of 0.25, 0.5, 0.75, and 1.0 g/cm³. The existing SARP's indicate that the array k_{eff} peaks with a film thickness between 0.5 and 5 cm thick. No overpack was modeled for these cases.

Selected results generated in this study are illustrated in Figures 6–8. The infinite array k_{eff} values are shown in Figure 6 for 7 wt.% ²³⁵U enrichment and in Figure 7 for 10 wt.% ²³⁵U enrichment. The results show that array k_{eff} peaks with a film thickness of about 1 cm (slightly less). Full density water was shown to be bounding in scoping calculations. Figure 8 shows the k_{eff} values for the infinite array with a 1 cm water thickness around the cylinder as a function of UF₆ enrichment and UF₆ density. The k_{eff} values presented in the figures are provided in Tables 4–6, and all the k_{eff} values for this study are provided in

Appendix A. Entries shaded in blue have k_{eff} values greater than 1.0, and those shaded in gray have k_{eff} values between 0.95 and 1.0. The Monte Carlo statistical uncertainty in the k_{eff} values is not presented in the tables, but all calculations in this study were run until an uncertainty of 0.05% Δk (50 pcm) was achieved.

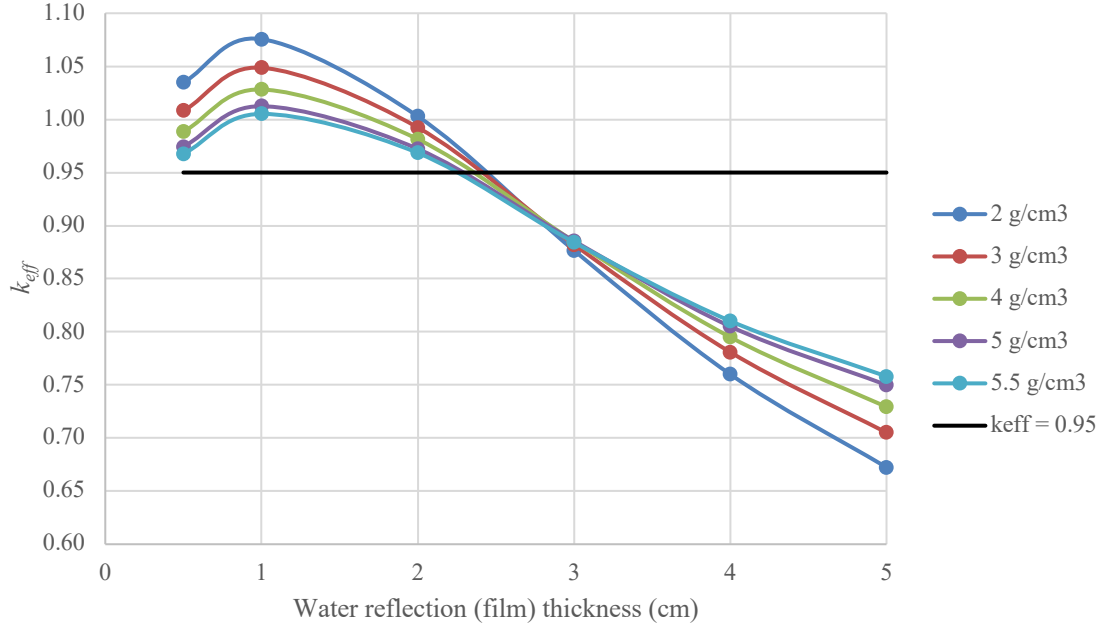


Figure 6. k_{eff} values as a function of water thickness and UF_6 density for 7 wt% ^{235}U enrichment.

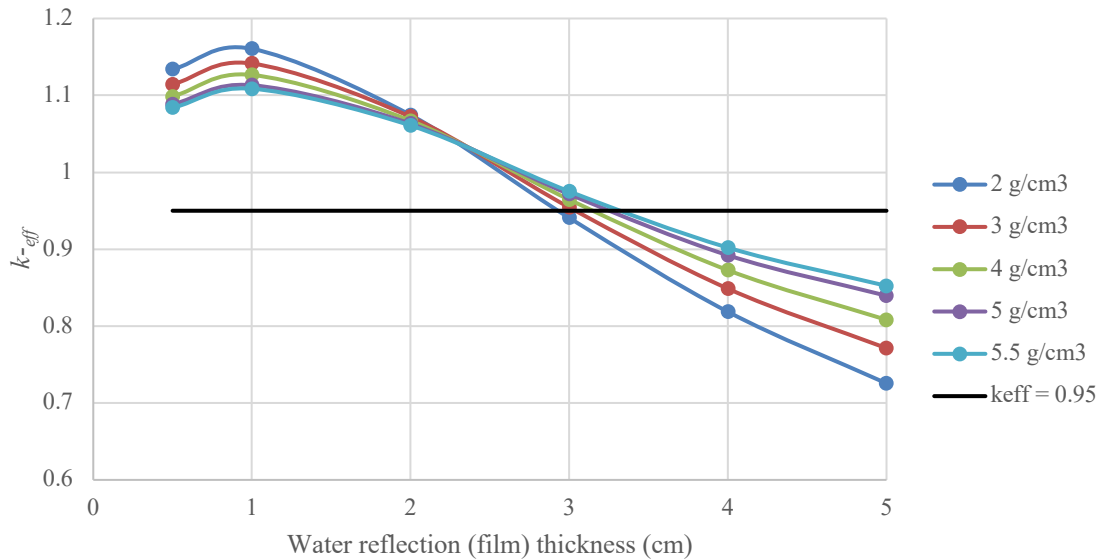


Figure 7. k_{eff} values as a function of water thickness and UF_6 density for 10 wt% ^{235}U enrichment.

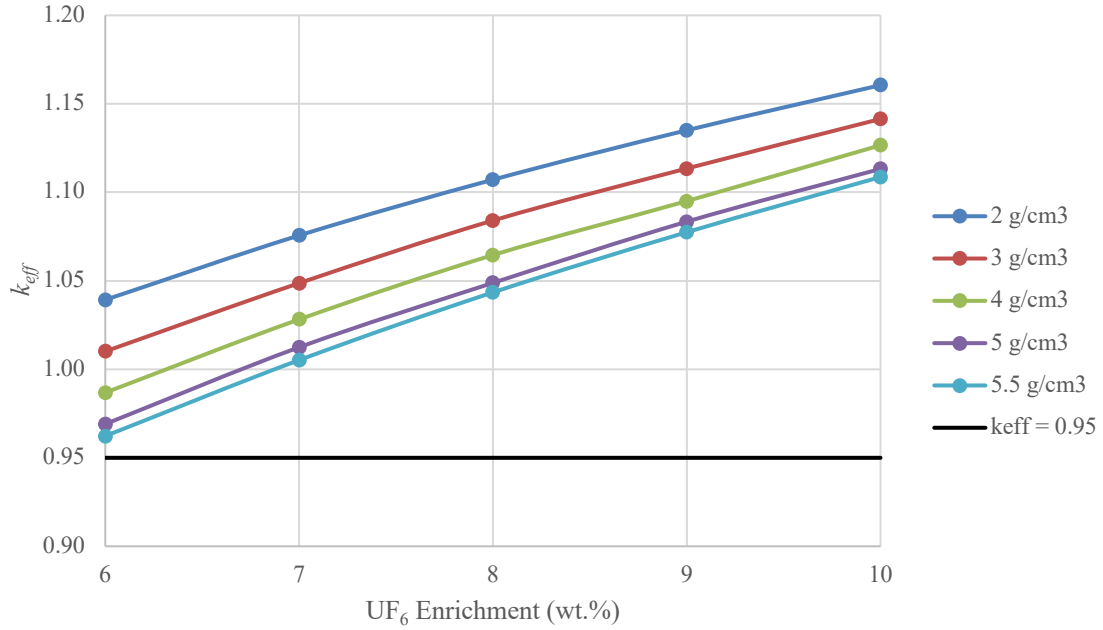


Figure 8. k_{eff} values as a function of ^{235}U and UF_6 density for a 1 cm water film around the cylinder.

Table 4. k_{eff} Values for infinite arrays with 7 wt.% ^{235}U enrichment

Film thickness (cm)	UF ₆ density (g/cm ³)				
	2	3	4	5	5.5
0.5	1.03490	1.00819	0.98850	0.97437	0.96748
1	1.07563	1.04867	1.02819	1.01244	1.00521
2	1.00291	0.99244	0.98138	0.97226	0.96849
3	0.87668	0.88217	0.88519	0.88555	0.88426
4	0.76002	0.78089	0.79502	0.80510	0.81024
5	0.67241	0.70511	0.72936	0.74995	0.75807

Table 5. k_{eff} Values for infinite arrays with 10 wt.% ^{235}U enrichment

Film thickness (cm)	UF ₆ density (g/cm ³)				
	2	3	4	5	5.5
0.5	1.13380	1.11438	1.09864	1.08837	1.08385
1	1.16055	1.14145	1.12660	1.11323	1.10866
2	1.07439	1.07211	1.06706	1.06316	1.06086
3	0.94075	0.95436	0.96451	0.97153	0.97519
4	0.81863	0.84869	0.87272	0.89213	0.90204
5	0.72553	0.77130	0.80817	0.83949	0.85235

Table 6. k_{eff} Values for infinite arrays with 1 cm water film around the cylinder

UF ₆ enrichment (wt.%)	UF ₆ density (g/cm ³)				
	2	3	4	5	5.5
6	1.03926	1.01013	0.98683	0.96907	0.96232
7	1.07563	1.04867	1.02819	1.01244	1.00521
8	1.10713	1.08406	1.06455	1.04881	1.04346
9	1.13504	1.11344	1.09504	1.08341	1.07746
10	1.16055	1.14145	1.12660	1.11323	1.10866

For the infinite arrays, k_{eff} increased with increasing UF_6 enrichment with k_{eff} peaking with a film thickness of about 1 cm (slightly less). Below a film thickness of 3 cm, the lower densities have higher k_{eff} values. The DN-30 and ORNL assessment contain similar results and show that with HUR present in the cylinders, k_{eff} increased with increasing density in the infinite arrays. These results indicate that higher enrichments will need to look at smaller, finite arrays in order to be considered acceptable under NCT or HAC.

3.4 FINITE ARRAY CALCULATIONS WITH 7 WT.% ^{235}U

The *Assessment of Existing Transportation Packages for Use with HALEU* (ORNL/TM-2020/1725) analysis used the HAC model for its array calculations since the DN-30 SARP determined the HAC model to be the bounding configuration (model with the collapsed overpack and the spherical HUR configuration). With this in mind, the studies documented here also start with the HAC model. To minimize the number of parameter studies, an N and an enrichment were determined for a starting configuration based on the ORNL/TM-2020/1725 results. The array size will then be increased or decreased as necessary to maximize acceptability under NCT and HAC. For the array size, N was chosen to be 7, with the HAC array being $2N$ (14), and the NCT array being $5N$ (35). A discussion of N and the required array sizes for NCT and HAC analyses is included above in Section 1.3. The starting enrichment was set at 7 wt.%.

3.4.1 Hypothetical Accident Condition Array

The HAC array of 14 cylinders was modeled as a 7×2 hexagonal array, as shown in Figure 9. For the HAC, the overpack is assumed to collapse into close proximity to the 30B cylinder wall (no foam material) for modeling purposes. The HUR sphere is included in the model, and the spheres are located within each cylinder to maximize interaction between the spheres. Axially, the spheres in the lower cylinders are pushed to the tops of the cylinders and the spheres in the upper cylinders are pushed to the bottoms of the cylinders. Radially, the outer cylinders are paired so that HUR spheres are directly adjacent to each other. The HUR sphere in the central cylinder is centered. The configuration of the HUR spheres is shown in Figure 10. The array was evaluated with the UF_6 density varying from 2.5 to 5.5 g/cm³ and the water film thickness around the 30B (inside the collapsed overpack) varying from 0 to 5 cm thick, with the overpack close fitting to the water.